

The constant 16777216 appears in `find_lossless_k()` and is used as a denominator (DENOM) to approximate the fractional part of π with 24-bit precision.

Why 16777216?

- $16777216 = 2^{24}$ – it is a power of two.
- The code computes:
 $k = \text{round}((\pi - 3) \times 2^{24})$

Then $\pi \approx 3 + \frac{k}{2^{24}}$.

- This yields a rational approximation of π with 24 bits of fractional resolution, enough to losslessly recover the first few decimal digits (tested for n=7 digits in the code).
- The associated bit-size logic (`get_bit_size`) uses 23 or 25 bits for k, confirming that the denominator is centered on 24 bits.
- Using a power of two avoids floating-point rounding issues and makes bit-mask generation straightforward.

In short: $16777216 = 2^{24}$ – a natural choice for a 24-bit fractional representation of π .